# Navneet Rishav

Machine Learning Engineer — GenAI & ML Systems
rishavnavneet@gmail.com — LinkedIn — GitHub — India

## Profile

Machine Learning Engineer with hands-on experience building, training, and deploying ML and GenAI solutions, including predictive models and LLM-based pipelines. Skilled in data preprocessing, feature engineering, model evaluation, and production deployment. Proven ability to translate business problems into scalable ML systems that improve decision-making, automation, and user outcomes.

## Experience

**AI Engineer** *June 2024 – Present*
*Regardo vision and sources, India*

- Designed, developed, and deployed machine learning and GenAI solutions across the full lifecycle, from data preprocessing to production APIs.
- Built and optimized ML pipelines for prediction, classification, and text analytics, improving model performance and inference efficiency.
- Implemented Retrieval-Augmented Generation (RAG) systems using embeddings and vector databases to enable accurate, domain-specific question answering.
- Developed RESTful APIs using FastAPI for real-time model inference and GenAI workflows, ensuring scalable and low-latency responses.
- Collaborated with cross-functional teams to translate business requirements into deployable ML systems and measurable outcomes.
- Applied best practices in model evaluation, error analysis, and prompt design to reduce hallucinations and improve output reliability.

## Technical Skills

- **Programming & Tools:** Python,C++, SQL, Linux, Git
- **Machine Learning:** Supervised & Unsupervised Learning, Time-Series Forecasting, NLP, Model Training & Evaluation
- **GenAI:** LLM Fundamentals,Embeddings & Vector Search, HuggingFace Transformers, Gradio, Prompt Engineering, RAG Architecture
- **Libraries & Frameworks:** Pytorch, Scikit-learn, LangChain, LangGraph, VectorDatabse, Matplotlib, OpenCV
- **Backend & Systems:** REST APIs, FastAPI, Django, Docker, AWS

## Projects

**GenAI Retrieval-Augmented Question Answering System**

- Designing a Retrieval-Augmented Generation (RAG) pipeline using embeddings and vector similarity search for domain-specific QA.
- Implementing document ingestion, chunking, embedding generation, and context-aware LLM inference.
- Evaluating retrieval relevance and response quality to reduce hallucinations and improve answer accuracy.

**Scalable Machine Learning Inference API**

- Building a production-style ML inference service using REST APIs for real-time predictions.
- Designing input validation, model loading workflows, and low-latency inference pipelines.
- Preparing the system for containerization and cloud deployment.

**LLM-Based Text Analytics & Insight Extraction System**

- Developing an LLM-powered system for summarization, classification, and insight extraction from unstructured text data.
- Comparing classical NLP pipelines with LLM-based approaches for performance and scalability.
- Designing prompt strategies to improve output consistency and factual correctness.

## Education

**Bachelor of Technology in Information Technology (CSE)** 2019 – 2023
Lovely Professional University, Punjab, India
Relevant Coursework: Data Structures, Algorithms, Operating Systems, Databases, Machine Learning, Artificial Intelligence

## Certifications

**LLM Engineering: Master AI & Large Language Models**
Udemy — Topics: Prompt Engineering, RAG Pipelines, LLM Agents, Vector Databases
Certificate ID: UC-29116926-6351-4aa9-a78b-af8e6fa92129

**Machine Learning & AI Foundations**
Udemy — Topics: Supervised Learning, NLP, Model Evaluation
Certificate ID: UC-ee880017-fa0f-430c-8e0f-cfd03ef904e4